Journal Name

Crossmark

RECEIVED
dd Month yyyy

REVISED
dd Month yyyy

PAPER

Few-Shot Classification of Lithium-Ion Battery Charging Power Curves with a Time-Series Foundation Model

Zhuotao Qian[1,*] and Chenghua Feng[1]

[1]Hangzhou Dianzi University Information Engineering School, Hangzhou 311305, Zhejiang, China *Author to whom any correspondence should be addressed.

E-mail: 20221009@hdu.edu.cn (Z.Q.); 239500314@hziee.edu.cn (C.F.)

Keywords: lithium-ion battery, charging power curve, time-series foundation model, few-shot learning, anomaly classification, mask-aware pooling, stacked ensemble

Abstract

Charging power curves from electric bicycles contain information associated with charger faults and battery anomalies, but verified fault labels are expensive to obtain. The records considered here also vary substantially in length, from 18 to 816 samples. Directly applying a general time-series foundation model therefore gives weak results, while a task-specific neural network becomes unstable when only a small labeled subset is available. We develop a three-class method that combines a frozen time-series foundation model with raw-scale dynamics features and a masked temporal convolutional network. First, mask-aware patch pooling prevents padded patches from entering the pooled MOMENT representation; this change raises the test Macro-F1 of a frozen linear probe from 69.46% to 77.41%. Second, a first-difference stream restores local changes that patch encoding tends to smooth, increasing the single-branch validation Macro-F1 from 83.91% to 93.57% when 40% of the training labels are used. A 60-dimensional dynamics branch retains absolute power levels removed by per-series normalization. The three branches are combined by five-fold cross-fitted probability stacking. Experiments on 35,099 operational charging records show validation Macro-F1 scores of $89.15 \pm 0.98\%$, $93.85 \pm 0.43\%$, and $95.02 \pm 0.33\%$ with 1%, 5%, and 10% of the training labels, respectively. At the 1% budget, the complete model exceeds the masked temporal convolutional network by 27.24 percentage points. With 40% labels, it reaches $98.09 \pm 0.23\%$ Macro-F1, 98.23% battery-anomaly recall, and a 0.67% false-positive rate. Same-architecture controls show that pretraining contributes 6.00–18.81 percentage points to the dual-stream branch. The results also identify clear trade-offs in inference cost and sensitivity to acquisition noise.

1 Introduction

Lithium-ion batteries power most electric bicycles used for short urban trips. During charging, an internal short circuit, cell mismatch, abnormal charger output, or failed protection circuit can lead to overheating and, in severe cases, thermal runaway. Detecting abnormal behavior during charging is therefore an important part of fire-risk reduction.

Many battery fault-diagnosis systems use cell voltage, temperature, current, and other signals collected by a battery management system (BMS) [1]. This information supports accurate diagnosis but is unavailable from many electric-bicycle batteries because models differ widely and external communication interfaces are often absent. A charging facility can usually collect only a single power sequence. If normal charging, battery anomalies, and charger faults can be distinguished from this sequence, the facility can provide non-intrusive monitoring without modifying the battery. Related studies have demonstrated non-intrusive fault identification for electric-vehicle charging equipment [2, 3], but electric-bicycle charging curves have received less attention.

This task presents three practical difficulties. First, record length varies from 18 to 816 samples. Batches require extensive right padding, and length itself is correlated with class. A model may consequently learn a padding shortcut instead of the power shape. Second, useful evidence occurs at different scales. Absolute power distinguishes low-power plateaus, local differences reveal spikes and steps, and the full curve describes the transition from a constant-power region to decay. Third,

Table 1. Dataset split and class distribution.

Subset	Normal (0)	Battery anomaly (1)	Charger fault (2)	Total
Training	8,480	7,014	9,075	24,569
Validation	1,212	1,002	1,296	3,510
Test	2,423	2,004	2,593	7,020
Total	12,115	10,020	12,964	35,099

fault labels require manual verification against maintenance records and are scarce in normal operations.

Time-series classifiers include feature-based methods [6, 7, 8], task-specific neural networks such as temporal convolutional networks (TCNs) [9] and Transformer variants [10, 11, 12], and random-kernel transforms such as ROCKET, MiniRocket, and MultiRocket [13, 14, 15]. Transfer learning and self-supervised representation learning can reduce label requirements [16, 17]. More recently, foundation models such as MOMENT, TimesFM, and Chronos have been pretrained on large, heterogeneous collections of time series [18, 19, 20]. Their representations have been studied in machinery diagnosis, engineering displacement prediction, and input-representation design [21, 22, 23].

Direct transfer is not sufficient in the present task. Freezing MOMENT-1-large and training its default classification head yields only 69.46% test Macro-F1, compared with 95.99% for a task-specific TCN trained on raw-scale curves. We trace this mismatch to three operations. The default head averages padded patches; patch encoding smooths short power changes; and per-series normalization removes absolute amplitude. We address each operation without updating the 341.24 million backbone parameters.

The contributions are as follows. First, mask-aware patch pooling excludes incomplete and padded patches and makes the frozen representation invariant to the padded region. Second, a first-difference stream supplies local change information to the same frozen encoder. Third, a 60-dimensional raw-scale dynamics branch restores amplitude and interpretable local statistics. Finally, five-fold cross-fitted stacking combines the foundation-model branch, the dynamics branch, and a masked TCN. Paired ablations, same-architecture pretraining controls, label-budget experiments, noise perturbations, and end-to-end latency measurements are used to distinguish the sources of improvement and their limits.

2 Problem formulation and data

2.1 Task definition

Let the valid power sequence for record i be $\mathbf{x}_i = (x_{i,1}, x_{i,2}, \ldots, x_{i,L_i})$, where $18 \le L_i \le 816$. For batching, the sequence is right-padded to $T = 1024$ and paired with a binary mask $\mathbf{m}_i \in \{0,1\}^T$, where $m_{i,t} = 1$ for a valid observation and $m_{i,t} = 0$ for padding. The classifier learns

$$f : (\mathbf{x}_i, \mathbf{m}_i) \mapsto y_i \in \{0, 1, 2\}, \tag{1}$$

where labels 0, 1, and 2 denote normal charging, a battery anomaly, and a charger fault, respectively. An ideal classifier is invariant to both the amount and the values of right padding.

2.2 Dataset

The data consist of operational records from an electric-bicycle charging platform. Each record contains an anonymized identifier, an operational label, and a charging power sequence. Of 46,535 original records, 11,435 carry the platform's label 5 for an invalid or incomplete record; their median length is one sample. These records were excluded under the platform's established protocol. Records shorter than 18 samples were also removed, leaving 35,099 valid sequences. The threshold of 18 was inherited from the existing experimental protocol; the data provider has not supplied an independent physical interpretation for it. Sampling intervals were not included in the source data, so this study reports sequence length only in samples and does not convert it to elapsed time.

No duplicate sequences or label conflicts were found. The longest sequence contains 816 samples and is not truncated at $T = 1024$. A stratified 7:1:2 split defines the training, validation, and test sets (table 1). Every method uses the same split files, which were checked by data hashes.

Figure 1 shows randomly selected training curves and the empirical cumulative distributions of valid length. The examples were selected with a fixed seed of 42 and were not curated. Normal curves often contain a stable power interval followed by smooth decay. Some battery-anomaly

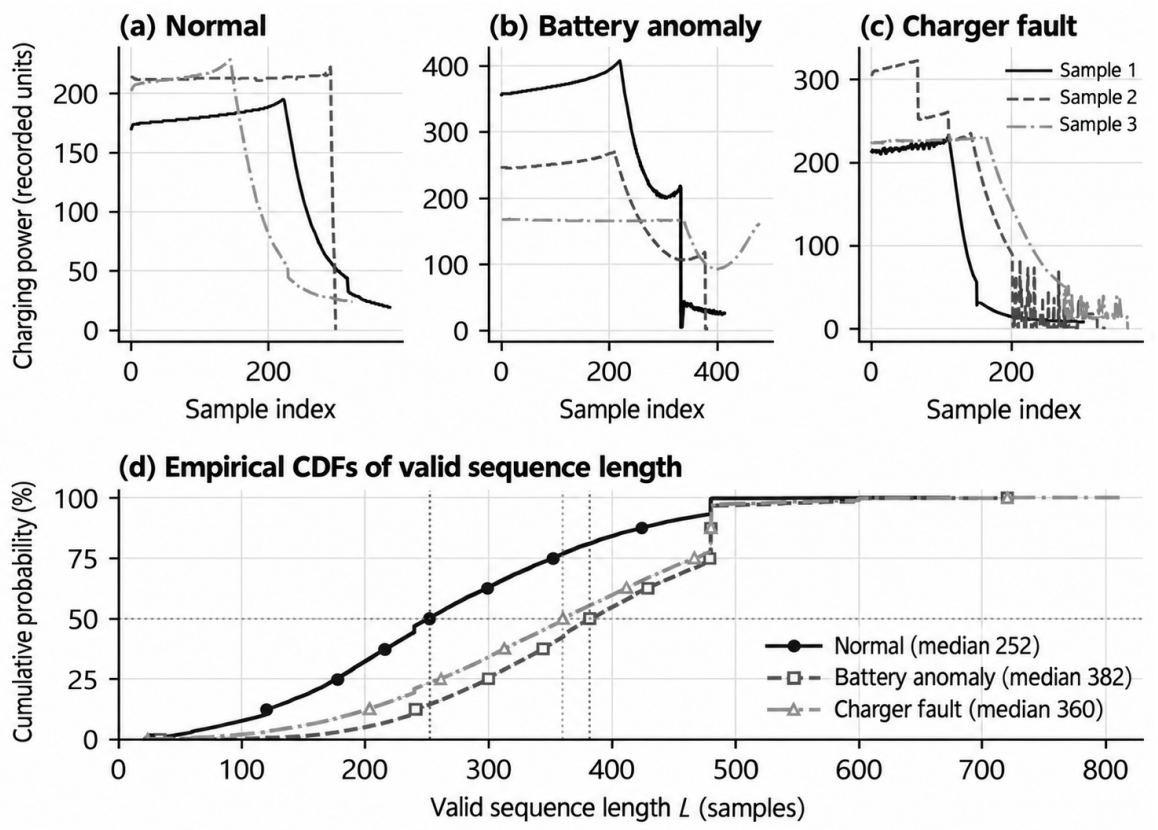


Figure 1. Examples and valid-length distributions in the training set. Panels (a)–(c) show three randomly selected sequences per class using seed 42. Panel (d) shows empirical cumulative distributions and class medians. The sharp rise near 480 samples reflects a common truncation point in the platform records.

examples contain abrupt drops, different plateau levels, or a late increase, while charger-fault examples contain step changes or repeated fluctuations during decay. These visual patterns describe the sampled records; they do not establish a physical failure mechanism.

The median valid lengths are 252, 382, and 360 samples for normal charging, battery anomalies, and charger faults, respectively. A depth-three decision tree trained on length alone obtains 35.76% test Macro-F1, compared with 17.98% for the majority-class baseline. Length is therefore a potential shortcut. All sequence models use masks, and section 4.6 also reports results by length group.

2.3 Evaluation metrics

The class counts are similar but not identical. The primary metric is the unweighted mean of the classwise F1 scores,

$$\text{Macro-F1} = \frac{1}{3}\sum_{c=0}^{2}\frac{2P_cR_c}{P_c+R_c}, \tag{2}$$

where P_c and R_c are precision and recall for class c. Because missing a battery anomaly is the most safety-relevant error, we also report recall for class 1 and its false-positive rate (FPR), defined as the proportion of non-battery-anomaly records predicted as class 1. Average precision is used for single-seed analyses because a precision-recall curve is more informative than an ROC curve for an imbalanced binary evaluation [26]. Probability quality is measured by expected calibration error (ECE) [27].

3 Method

3.1 Overview

The framework in figure 2 has three branches. Branch I sends the original and first-difference sequences through a shared frozen MOMENT-1-large encoder and uses mask-aware pooling. Branch II extracts 60 raw-scale dynamics features and classifies them with a histogram gradient-boosting model (HGB). Branch III applies a six-layer causal dilated TCN and masked mean pooling. Each

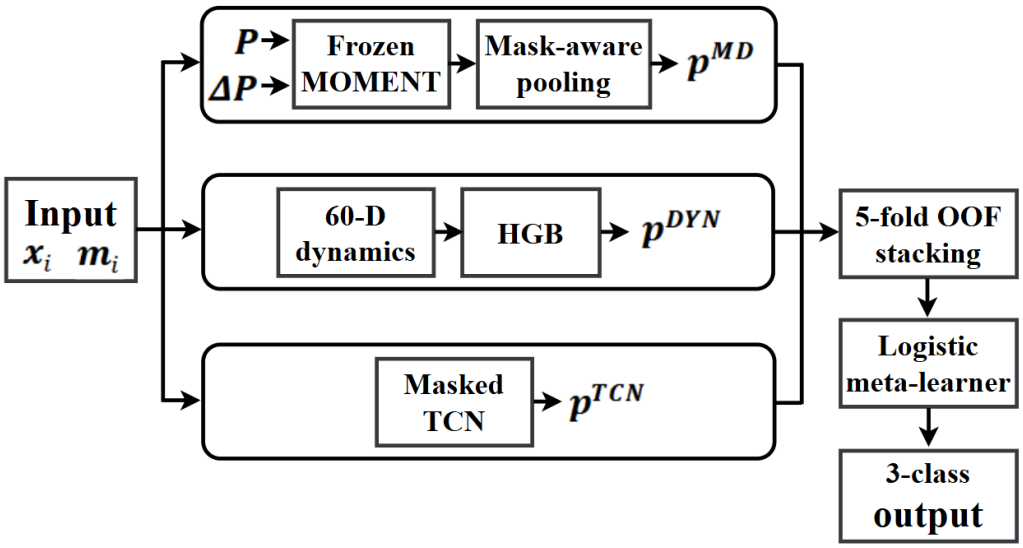


Figure 2. Proposed three-branch framework. The original power sequence P and its first difference ΔP share a frozen MOMENT encoder. The other branches use raw-scale dynamics features and a masked TCN. Five-fold OOF probabilities train the logistic meta-learner.

branch produces three class probabilities. Five-fold out-of-fold (OOF) predictions form a nine-dimensional meta-feature for multinomial logistic regression.

3.2 Mask-aware patch pooling

MOMENT-1-large uses the T5-large encoder architecture with 24 Transformer layers, a hidden size of 1024, and 16 attention heads. It contains 341.24 million parameters. Similar to PatchTST [24], it divides an input of length T into $N = T/P = 128$ non-overlapping patches of size $P = 8$, producing representations $\mathbf{h}_{i,1}, \ldots, \mathbf{h}_{i,N} \in \mathbb{R}^{1024}$.

In momentfm 0.1.4, the default classification head passes the input mask into the encoder but averages all patch representations:

$$\mathbf{z}_i^{\text{default}} = \frac{1}{N} \sum_{n=1}^{N} \mathbf{h}_{i,n}. \tag{3}$$

For a sequence of length 18, only two of 128 patches are complete; the remaining padded patches dominate the average. By contrast, approximately 78% of the patches are valid for a sequence of length 800. The degree of representation dilution therefore depends on length.

We define a patch-validity mask

$$\widetilde{m}_{i,n} = \prod_{t=(n-1)P+1}^{nP} m_{i,t}, \qquad n = 1, \ldots, N, \tag{4}$$

and pool only complete patches:

$$\mathbf{z}_i = \frac{\sum_{n=1}^{N} \widetilde{m}_{i,n} \mathbf{h}_{i,n}}{\sum_{n=1}^{N} \widetilde{m}_{i,n}}. \tag{5}$$

Because $L_i \geq 18$, at least two patches are complete and the denominator is nonzero. On fixed synthetic sequences with lengths from 18 to 384, changing the padded length from 512 to 1024 or replacing masked values by random values changes the pooled representation by at most 4.2×10^{-7}, which is at the scale of single-precision rounding. This invariance applies to the padded region; external z-score normalization and the model's reversible instance normalization (RevIN) [25] still use valid observations in an incomplete final patch.

The backbone is frozen, so the representations for all data splits are computed once and cached. Downstream training no longer repeats the 341.24-million-parameter forward pass. One classifier epoch consequently falls from approximately 2 min to approximately 2 s.

3.3 First-difference dual-stream representation

Patch encoding and instance normalization favor global shape but can smooth the spikes, slope changes, and plateau transitions that distinguish the classes. Full fine-tuning can recover local

Table 2. Groups in the 60-dimensional raw-scale dynamics feature vector.

Group	Dim.	Representative features	Interpretation
Amplitude and quantiles	17	Mean, standard deviation, extrema, 5th–95th percentiles, coefficient of variation, end/start ratio	Power level and distribution
Roughness and turning points	12	Absolute-difference statistics, upward fraction, direction-change rate, plateau fraction, longest plateau	Local fluctuation and jump frequency
Segment trends	18	Global and segment slopes, segment means and deviations, final-minus-initial segment	Constant-power and decay behavior
Spectrum and autocorrelation	7	Four band-energy ratios, dominant frequency, lag autocorrelation, histogram entropy	Periodicity and complexity
Shape	4	Maximum drawdown, maximum rise, skewness, kurtosis	Abrupt change and distribution shape
Length	2	Valid length and fraction of T	Number of valid samples

information, but it updates all backbone parameters, takes about 3 h per run in our setting, and can overfit at low label budgets. Instead, we expose local change as a second input. Within the valid interval,

$$\Delta x_{i,t} = x_{i,t+1} - x_{i,t}, \qquad t = 1, \ldots, L_i - 1. \tag{6}$$

Slow trends move toward zero, whereas abrupt changes become prominent impulses. The original and difference sequences are z-score normalized separately, passed through the same frozen encoder, and pooled using equation (5). Their concatenation is

$$\mathbf{z}_i^{\mathrm{MD}} = [\mathbf{z}_i^{P} \,\|\, \mathbf{z}_i^{\Delta P}] \in \mathbb{R}^{2048}. \tag{7}$$

An L2-regularized multinomial logistic regression with $C = 1.0$ maps this vector to $\mathbf{p}_i^{\mathrm{MD}}$. The additional cost is one frozen-encoder forward pass per record; no new backbone parameter is trained.

3.4 Raw-scale dynamics features

Per-series z-score normalization and RevIN remove absolute power. This is undesirable when a low-power plateau and a normal constant-power interval have similar normalized shapes. In an early control, a raw-scale TCN obtained 95.99% test Macro-F1, whereas per-series min-max normalization reduced it to 92.00%.

The Dyn branch therefore extracts 60 features from valid observations at the original scale. Table 2 summarizes the six groups. Fifteen dimensions are standard global statistics; the remaining 45 describe local dynamics, frequency content, and shape. Padding is never used. A fixed HGB classifier uses 400 maximum iterations, learning rate 0.08, 31 maximum leaves, 20 minimum samples per leaf, and L2 regularization 1.0 for every label budget. Feature extraction and HGB training take only seconds on a CPU at the 40% budget.

For example, the direction-change rate is the fraction of adjacent nonzero differences whose signs reverse; it captures repeated fluctuations. The plateau fraction is the proportion of adjacent samples whose absolute difference is below 10^{-3}.

3.5 Masked temporal convolutional branch

The TCN follows the platform's six-layer causal dilated architecture: kernel size $k = 3$, dilations $d_l = 2^{l-1}$ for $l = 1, \ldots, 6$, 128 channels per layer, ReLU activation, and SpatialDropout of 0.3. Its

receptive field is

$$R = 1 + (k-1)\sum_{l=1}^{6} d_l = 1 + 2 \times 63 = 127. \tag{8}$$

The original platform network globally averaged the full padded tensor. We replace this operation with

$$\mathbf{g}_i = \frac{\sum_{t=1}^{T} m_{i,t}\mathbf{c}_{i,t}}{\sum_{t=1}^{T} m_{i,t}}, \tag{9}$$

where $\mathbf{c}_{i,t}$ is the final convolutional representation. Causality ensures that a valid position does not depend on right padding, and equation (9) excludes padded outputs. A dense layer and softmax produce $\mathbf{p}_i^{\mathrm{TCN}}$.

The TCN uses raw-scale inputs and AdamW with learning rate 10^{-3}, weight decay 10^{-4}, batch size 32, gradient clipping at 1.0, and a 50-epoch cosine schedule. The epoch-50 endpoint is used in fusion experiments to avoid choosing an epoch on the validation set. Training uses FP32 because FP16 dilated convolutions showed late-epoch overflow on the V100 GPU.

3.6 Cross-fitted probability stacking

Training a meta-learner on in-sample predictions would favor overfitted branches. We instead divide each labeled subset $\mathcal{D}$ into five stratified folds. For fold k, all three branches train on $\mathcal{D} \setminus \mathcal{D}_k$ and predict $\mathcal{D}_k$. Concatenated OOF probabilities form

$$\mathbf{u}_i = [\mathbf{p}_i^{\mathrm{TCN}}, \mathbf{p}_i^{\mathrm{DYN}}, \mathbf{p}_i^{\mathrm{MD}}] \in [0,1]^9. \tag{10}$$

An L2-regularized multinomial logistic regression estimates

$$P(y_i = c \mid \mathbf{u}_i) = \frac{\exp(\mathbf{w}_c^\top \mathbf{u}_i + b_c)}{\sum_{c'=0}^{2} \exp(\mathbf{w}_{c'}^\top \mathbf{u}_i + b_{c'})}. \tag{11}$$

Its regularization is selected by an inner three-fold split of the OOF features. Each branch is then retrained on the complete labeled subset for inference. Neither the validation nor the test set enters fitting or hyperparameter selection. The meta-learner has only 30 parameters. When calibrated probabilities are required, a scalar temperature can be fitted on the OOF logits by minimizing negative log likelihood.

4 Experiments and results

4.1 Experimental protocol

Label budgets of 1%, 5%, 10%, 20%, 40%, and 100% are sampled stratifiably from the training split, corresponding to 244, 1,227, 2,456, 4,913, 9,827, and 24,569 records. Within a seed, every method uses the same labeled subset and evaluation records. Unless stated otherwise, results are the mean and sample standard deviation over seeds 42, 44, 45, and 46. Method comparisons use paired seed differences and report descriptive 95% paired t intervals. Exact McNemar tests are used for single-seed error-correction analyses.

The study has two stages. The first stage uses five independent test-set runs to evaluate frozen-MOMENT pooling, adaptation, and pretraining controls. After the test set had been inspected, all further method development and ablation used only the validation set; the test set was not reopened. The four validation splits overlap across seeds and were used repeatedly during development, so paired intervals describe uncertainty within this protocol rather than external generalization. New data are required for a final independent evaluation.

Experiments use Python 3.11, PyTorch 2.12.1 with CUDA 12.6, momentfm 0.1.4, and scikit-learn. MOMENT feature extraction and network training use an NVIDIA Tesla V100-PCIE 32 GB. End-to-end latency is measured on an RTX 4080, and the unified label-budget reruns use an RTX 5090.

4.2 Baselines and initial frozen-model performance

Table 3 reports five-seed test results with all training labels. A raw-scale TCN reaches 95.99% Macro-F1, and a random forest on 15 statistics reaches 88.89%. The frozen default MOMENT head yields 69.46%. Mask-aware pooling alone adds 7.95 percentage points. An RBF-SVM on the corrected representation reaches 85.17%, while full fine-tuning reaches 95.37% at far higher training cost.

Table 3. Test performance with all training labels. Values are mean ± sample standard deviation over five seeds unless noted.

Method	Trainable parameters	Accuracy (%)	Macro-F1 (%)
Majority class	–	36.94 ± 0.00	17.98 ± 0.00
F15 + logistic regression	–	74.13 ± 0.59	73.75 ± 0.60
F15 + random forest	–	88.85 ± 0.23	88.89 ± 0.23
1D-CNN, per-series z-score	0.05 M	94.29 ± 1.27	94.38 ± 1.27
TCN, per-series min-max	0.22 M	91.79 ± 1.37	92.00 ± 1.37
TCN, per-series z-score	0.22 M	94.92 ± 1.44	95.02 ± 1.43
TCN, raw scale	0.22 M	95.94 ± 0.74	95.99 ± 0.73
Frozen MOMENT + linear head, default pooling[a]	3,075	69.53 ± 0.78	69.46 ± 0.76
Frozen MOMENT + linear head, masked pooling	3,075	77.02 ± 0.88	77.41 ± 0.89
Frozen MOMENT + RBF-SVM, masked pooling	0 backbone	84.77 ± 1.04	85.17 ± 1.03
MOMENT, final two layers unfrozen	25.70 M	81.08 ± 1.13	81.50 ± 1.06
Fully fine-tuned MOMENT	341.24 M	95.26 ± 0.41	95.37 ± 0.39

[a]Four seeds completed. The RBF-SVM follows the MOMENT protocol and selects its parameters by five-fold validation on at most 10,000 training records.

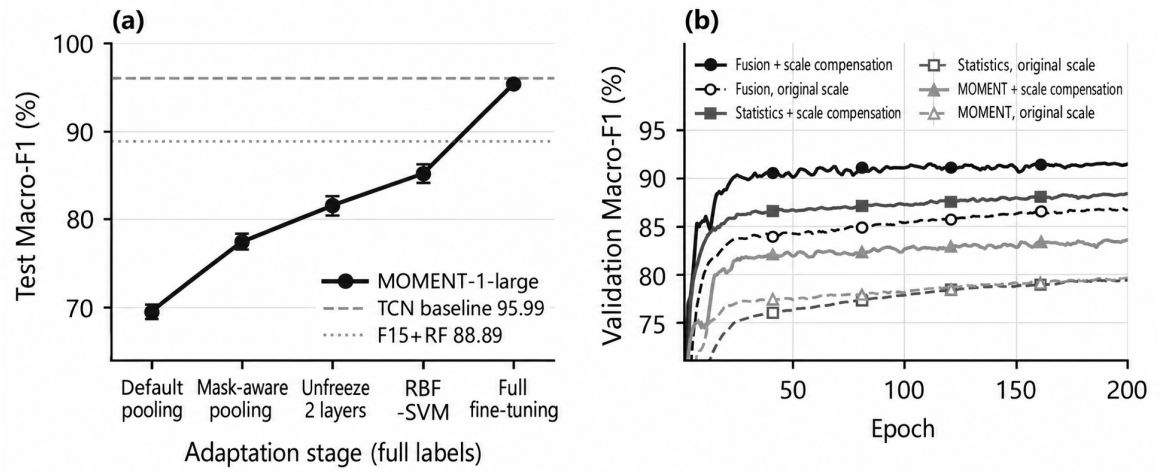


Figure 3. Effects of pooling and downstream optimization. (a) Test Macro-F1 along the MOMENT adaptation sequence. Error bars show seed standard deviations. (b) Validation convergence of frozen-feature heads, with and without fixed input-scale compensation.

Figure 3(a) summarizes the adaptation sequence. Full fine-tuning approaches the TCN but updates the complete backbone and takes approximately 3 h per run. Frozen feature extraction takes approximately 106 s once, after which downstream models train on a CPU.

Figure 3(b) exposes a separate optimization issue. When frozen representations and statistics are concatenated and sent to an MLP head, the initial activation root mean square (RMS) is only 0.024–0.029. Multiplying the input by $\gamma = 10^{3/9}/B$, where B is the number of active branches, restores an RMS of 0.49–0.55. At epoch 30 this scale correction adds 6.58 percentage points, and the best validation Macro-F1 within 200 epochs rises from 86.94% to 91.72%. Across four test seeds, the corrected head reaches $91.52 \pm 0.38\%$, exceeding an RBF-SVM on the same features by 0.56 points (95% interval 0.52–0.61). A frozen-model comparison is therefore meaningful only when the downstream head has been adequately optimized.

4.3 Padding sensitivity and TCN restoration

The platform's original globally averaged TCN also depends on padding. With weights fixed and valid observations unchanged, increasing total length from 1024 to 1536 changes the largest logit by 418–531 and changes predictions for 100 validation records at the 10% budget and 33 at the 100% budget. Masked pooling reduces the measured logit change to zero.

The restored six-layer architecture is also important. An early four-block residual reconstruction with receptive field 61 reaches only $69.23 \pm 1.61\%$ at 10% labels. The six-layer network with receptive field 127 reaches $93.68 \pm 1.93\%$, $97.78 \pm 0.16\%$, and $98.40 \pm 0.17\%$ at 10%, 40%, and 100%, respectively. At 1% and 5%, however, it falls to $55.51 \pm 14.66\%$ and $84.03 \pm 4.57\%$,

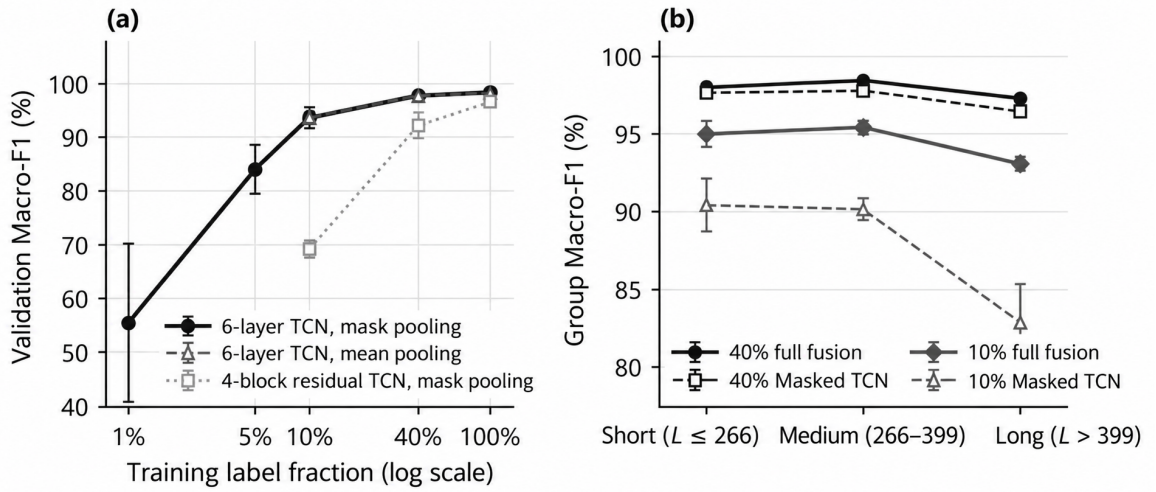


Figure 4. TCN architecture, pooling, and length-stratified performance. (a) Validation Macro-F1 for three TCN configurations. (b) Performance by seed-specific training-length terciles. Long sequences exceed the TCN receptive field and show the largest fusion gain at the 10% label budget.

Table 4. Same-architecture comparison of pretrained and randomly initialized encoders. Macro-F1 values are percentages.

Evaluation	Classifier	Labels	Input	Pretrained	Random	Difference (pp)
Test, five seeds	RBF-SVM	1%	Original	68.05 ± 1.59	39.44 ± 5.00	+28.60
Test, five seeds	RBF-SVM	5%	Original	77.40 ± 0.93	54.81 ± 0.95	+22.58
Test, five seeds	RBF-SVM	10%	Original	80.77 ± 0.59	58.76 ± 1.26	+22.01
Validation, four seeds	Logistic	1%	Original	68.97 ± 1.55	46.20 ± 0.68	+22.77
Validation, four seeds	Logistic	1%	Difference	80.57 ± 0.63	64.99 ± 1.82	+15.57
Validation, four seeds	Logistic	1%	Dual	83.76 ± 1.99	64.95 ± 1.60	+18.81
Validation, four seeds	Logistic	10%	Original	81.27 ± 0.35	58.79 ± 1.13	+22.48
Validation, four seeds	Logistic	10%	Difference	88.31 ± 0.84	78.94 ± 1.07	+9.37
Validation, four seeds	Logistic	10%	Dual	91.24 ± 0.57	81.27 ± 0.91	+9.97
Validation, four seeds	Logistic	40%	Original	84.02 ± 0.53	63.45 ± 1.14	+20.57
Validation, four seeds	Logistic	40%	Difference	89.20 ± 0.51	83.03 ± 1.02	+6.17
Validation, four seeds	Logistic	40%	Dual	92.77 ± 0.58	86.77 ± 0.69	+6.00

revealing strong seed sensitivity. Subsequent experiments use this restored masked six-layer TCN as the baseline and ensemble branch.

4.4 Difference stream and contribution of pretraining

Figure 5(a) compares frozen MOMENT input streams. The difference stream alone exceeds the original stream. Their concatenation adds 13.38, 10.03, and 9.66 percentage points over the original stream at 1%, 10%, and 40% labels, respectively, with the same direction in all four seeds. At 40%, battery-anomaly recall rises from 89.42% to 93.36% and its FPR falls from 3.81% to 2.56%. Charger-fault recall rises from 77.87% to 94.31%, while its FPR falls from 9.60% to 3.43%. Stacking the dual stream with DYN reaches 88.69% at 1% labels.

To separate pretraining from encoder capacity, we replace pretrained weights by random initialization while holding the architecture, masks, preprocessing, and downstream classifier fixed. Table 4 shows that pretrained weights win in every reported condition and seed. For the dual stream, the advantage is 18.81, 9.97, and 6.00 percentage points at 1%, 10%, and 40%, respectively. The benefit increases as labels become scarcer. The difference stream also helps a random encoder, so local differencing and pretrained priors make distinct, additive contributions.

The three test differences have paired 95% intervals of 22.24–34.97, 20.89–24.28, and 20.48–23.53 points. On validation data, the dual-stream pretraining advantages at 1%, 5%, 10%, 20%, and 40% labels are 18.81, 12.51, 9.97, 7.57, and 6.00 points. Figure 5(b) shows this trend.

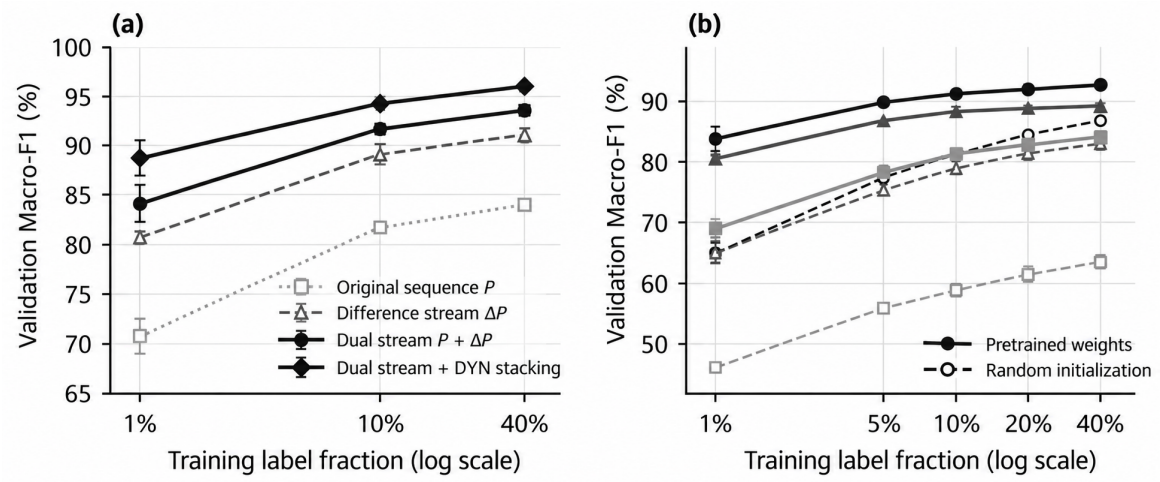


Figure 5. Difference-stream and pretraining effects. (a) Validation Macro-F1 for original, difference, and dual-stream frozen MOMENT representations, with a DYN stack. (b) Same-architecture comparison of pretrained and randomly initialized weights.

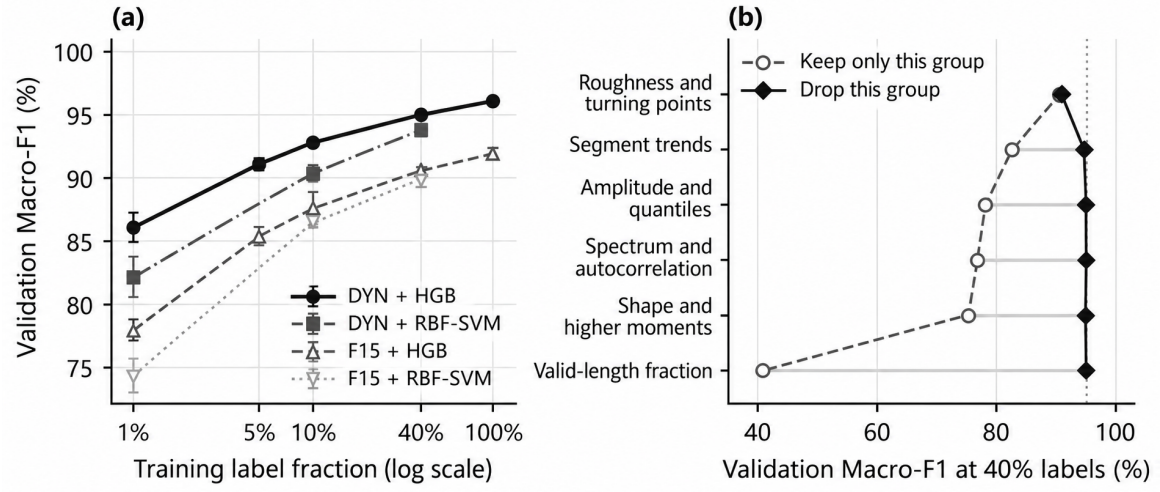


Figure 6. Raw-scale dynamics features. (a) Label-budget performance of F15 and DYN with HGB and RBF-SVM. (b) Grouped keep-one and drop-one analysis at 40% labels. The vertical dotted line is the complete DYN result.

4.5 Dynamics features

Figure 6(a) compares the 60 DYN features with the 15 global statistics (F15). DYN is better at every label budget, and HGB is better than RBF-SVM. At 40% labels, DYN+HGB reaches $95.00 \pm 0.17\%$, 4.40 points above F15+HGB. At 1%, it reaches 86.05%, 8.12 points above F15+HGB and about two points above the frozen dual-stream branch.

The grouped analysis in figure 6(b) identifies roughness and turning points as the most important group. Removing it reduces the 40% result by 4.13 points, whereas using it alone reaches 90.48%. Segment trends are the next most informative group. Removing amplitude, spectrum, or shape individually has little effect, but each group alone still reaches approximately 75–78%, indicating redundancy. Length alone reaches 40.81%, confirming that the branch relies mainly on power shape rather than duration.

4.6 Fusion, complementarity, and label budgets

At the 10% budget, the TCN obtains 89.04% Macro-F1 and 93.11% battery-anomaly recall. Full fusion raises recall to 95.28% and lowers the FPR from 2.89% to 1.87%; TCN+DYN has marginally higher recall at 95.33%. At 40%, full fusion increases battery-anomaly recall from 97.83% for the TCN to 98.23% and reduces FPR from 0.77% to 0.67%. It exceeds TCN+DYN by only about 0.05 points in recall.

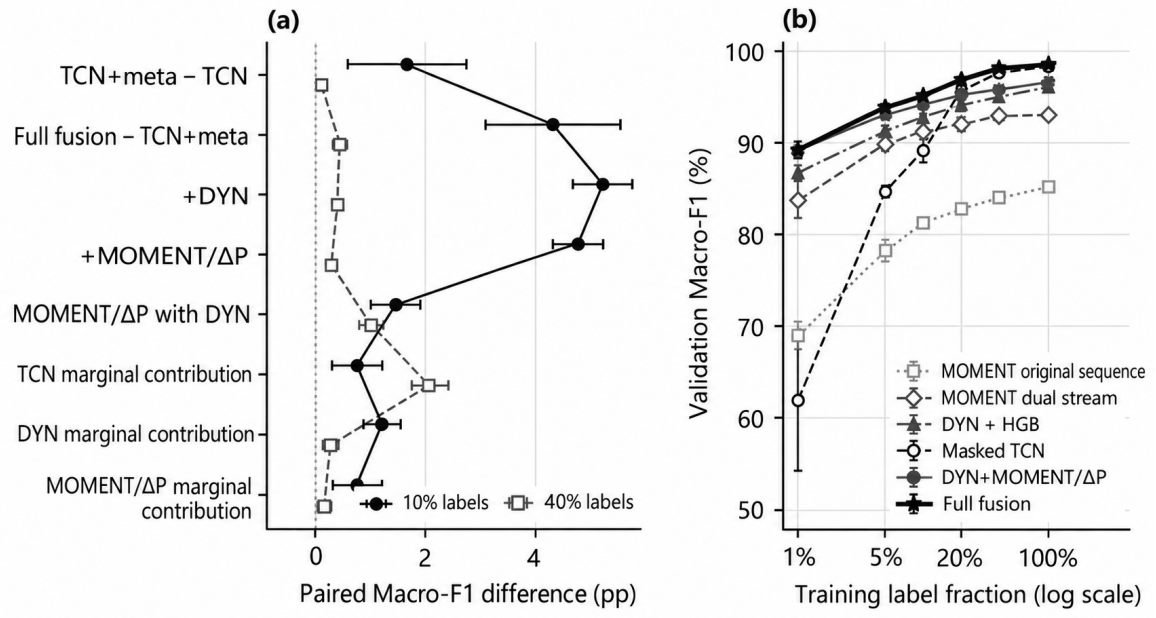


Figure 7. Complementarity and label-budget effects. (a) Paired branch and same-form meta-learner comparisons at 10% and 40% labels. Error bars show seed standard deviations. (b) Validation Macro-F1 under the unified fixed-hyperparameter protocol.

Figure 7(a) separates the information gain from the effect of another classifier layer. A meta-learner fed only TCN probabilities gives little improvement. Full fusion exceeds this same-form control in all four seeds. Removing TCN from the full model reduces Macro-F1 by 0.76 points at 10% and 2.07 points at 40%; removing Dyn reduces it by 1.21 and 0.27 points; removing the Moment/difference branch reduces it by 0.76 and 0.16 points. TCN contributes most at the higher budget, whereas Dyn and Moment contribute more when labels are scarce. Adding Moment to TCN+Dyn is consistently useful at 10%, but at 40% its small mean increment is positive in three of four seeds and its interval crosses zero.

At 40%, full fusion corrects 23, 26, 28, and 33 TCN errors across the four seeds while introducing 9, 7, 6, and 7 new errors. Net correction ranges from 14 to 26 records; exact McNemar p values are 0.020, 0.0013, 0.0002, and below 10^{-4}. Figure 4(b) shows the strongest gain for long sequences. At 10% labels, full fusion raises the long-sequence Macro-F1 from 82.90% to 93.07%, a 10.17-point improvement. Long records exceed the TCN receptive field of 127, whereas global attention directly connects distant patches.

At 40%, the fused model has an ECE of 0.56%. At 10%, it is under-confident: mean confidence is 92.21% while accuracy is 95.01%, giving an ECE of 3.52%. OOF temperature scaling lowers ECE to 0.94% and negative log likelihood from 0.1717 to 0.1518. A descriptive validation-set threshold scan for class 1 gives 97.38% recall at an observed 0.52% FPR and 98.75% recall at a 1.04% FPR for the 40% model. Because the threshold is selected and evaluated on the same validation data, deployment thresholds must be fixed on a separate calibration set.

Figure 7(b) summarizes the common fixed-hyperparameter label-budget runs. At 1%, the TCN averages 61.91% with substantial seed variation, while adding the Moment/difference branch raises TCN+Moment to 84.21% and full fusion to 89.15 ± 0.98%. Full fusion reaches 93.85 ± 0.43% at 5% and 95.02 ± 0.33% at 10%. Relative to TCN+Dyn, its gains at 1%, 5%, and 10% are 2.35, 1.53, and 0.77 points, with descriptive paired intervals excluding zero. At 20–40%, the increment over TCN+Dyn shrinks to approximately 0.06–0.14 points. With all labels, TCN+Dyn and full fusion reach 98.54% and 98.53%, respectively. MultiRocket is 1.03–3.37 points better than full fusion at 1–10%, ties it near 20% (96.92% versus 96.90%), and is 0.90 points lower at 40% (97.25% versus 98.14%). The proposed model is therefore not uniformly best among all time-series classifiers; its main value is the adaptation and complementary use of a frozen foundation model.

4.7 Inference cost and noise sensitivity

Table 5 reports end-to-end inference on one RTX 4080 for the 40% models. Timing includes preprocessing, Dyn extraction, two frozen-Moment passes, the TCN, and all classifier heads, but

Table 5. End-to-end inference latency on an RTX 4080 for models trained with 40% of the labels. Values are median milliseconds per record.

Method	Macro-F1 (%)	Batch 1	Batch 32	Batch 128
TCN	97.53	0.88	0.039	0.050
Dyn + HGB	95.00	11.43	1.99	1.83
TCN + Dyn	97.93	12.60	2.09	1.93
Full fusion	98.09	46.10	11.13	11.42

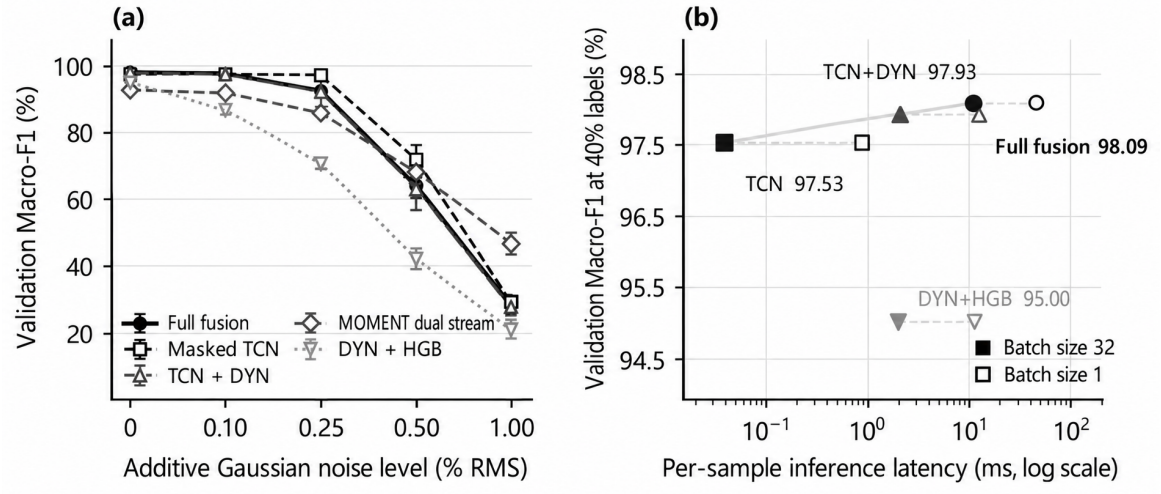


Figure 8. Deployment-related analyses at 40% labels. (a) Validation Macro-F1 after additive Gaussian noise is applied to valid observations. (b) Accuracy-latency trade-off for batch sizes 1 and 32 on an RTX 4080.

excludes disk and serving overhead. At batch size 32, full fusion takes 11.13 ms per record, corresponding to about 90 records per second, with 2.49 GB peak GPU memory. At batch size 1, the two Moment passes take approximately 33.4 ms of the 46.10 ms total. The stored FP32 Moment backbone is 1,302 MB, compared with 0.9 MB for the TCN, 4.2 MB for HGB, and less than 1 kB for the meta-learner. TCN+Dyn is a practical lower-cost operating point when labels are abundant.

For a noise sensitivity check, zero-mean Gaussian noise is added only to valid observations. Its standard deviation is 0.1%, 0.25%, 0.5%, or 1% of the training-set valid-observation RMS, which is 192.31 in recorded units. All models are trained on clean data without augmentation. As figure 8(a) shows, full fusion remains at 97.72% under 0.1% noise and falls to 92.49% at 0.25%. At 0.5%, it falls to 64.00%, while the TCN remains at 72.00%. The Dyn branch is particularly sensitive because its maximum-rise feature divides by a running minimum; values near zero can amplify a small perturbation. The frozen dual stream degrades more slowly, retaining 85.89%, 68.23%, and 46.64% at 0.25%, 0.5%, and 1% noise. Quantization shows a similar pattern: full fusion reaches 95.44% at a two-unit step and 64.73% at a five-unit step.

The risk metrics change in an unfavorable direction under stronger noise. For full fusion, battery-anomaly recall is 96.71%, 73.28%, and 14.10% at 0.25%, 0.5%, and 1% noise, while its FPR falls to 0.57%, 0.19%, and 0.04%. The lower FPR reflects missed battery anomalies, not improved robustness. Charger-fault FPR simultaneously rises to 11.02%, 53.21%, and 92.42%. At a five-unit quantization step, battery-anomaly recall is 68.01% with a 0.25% FPR. These perturbation results identify a failure boundary; they do not demonstrate deployment robustness.

5 Discussion

5.1 Role of the foundation model

The foundation model requires task-specific adaptation. Without corrected pooling, its frozen representation performs worse than a random forest on simple statistics. Mask-aware pooling removes padding dilution, the first-difference stream restores short changes, and Dyn restores raw amplitude. Same-architecture controls show that encoder capacity alone does not explain the final representation: pretrained weights add 6.00–18.81 points to the dual stream across 1–40% label budgets. The benefit is largest when labels are scarce.

The branches are complementary rather than interchangeable. Moment supplies cross-domain shape priors and global context; Dyn supplies interpretable amplitude and roughness; and the TCN learns fine local patterns when enough labels are available. Cross-fitted probabilities let the meta-learner use those differences without training on in-sample confidence. The length-stratified and noise experiments also show where the foundation representation adds the most: long sequences and moderate perturbations.

5.2 Comparison with alternative adaptation strategies

Fully fine-tuning Moment reaches 95.37% with all labels but updates 341.24 million parameters and takes about 3 h per run. The frozen approach needs one feature extraction pass, after which its downstream models train in minutes. These figures come from different label budgets and evaluation stages and should not be treated as a direct accuracy ranking. A rank-eight LoRA control on the attention projections adds only 0.45 points over a well-trained frozen fusion head across four seeds while increasing training cost by more than an order of magnitude. The frozen design was retained on cost-benefit grounds.

MultiRocket remains competitive and is better in the most label-scarce conditions. This finding narrows the claim of the present work: the contribution is an analysis of how to adapt a time-series foundation model to variable-length charging curves and combine it with other representations, not a claim of universal superiority. A future four-branch ensemble could add random-kernel features.

5.3 Limitations

The data come from one operating platform and lack device model, site, calendar time, and sampling-interval metadata. Cross-site, cross-period, and cross-device generalization has not been established. The second-stage ablations repeatedly use overlapping validation splits, so a new external test set is required before deployment. All models train on clean data and are sensitive to noise at approximately one recorded unit or above; some extreme-ratio Dyn features are numerically unstable. Finally, the operational labels are coarse. Class 1 denotes a battery anomaly and class 2 a charger fault under the provider's convention, but detailed failure modes and the provider's original label specification were not released. Curves and feature importance establish statistical associations, not battery-health mechanisms.

6 Conclusion

This study analyzes why a general time-series foundation model transfers poorly to variable-length electric-bicycle charging curves and proposes three targeted corrections. Mask-aware patch pooling raises the frozen linear-probe test Macro-F1 from 69.46% to 77.41% and removes dependence on right padding. A first-difference stream adds 9.66–13.38 points to the frozen branch without training the backbone. A 60-dimensional raw-scale dynamics branch restores amplitude information, reaching 95.00% Macro-F1 by itself at the 40% budget. Cross-fitted stacking reaches 95.02% at 10% labels and 98.09% at 40%, with 98.23% battery-anomaly recall and a 0.67% FPR at the latter budget. At 1% labels, it reaches 89.15%, 27.24 points above the masked TCN. The additional foundation-model benefit becomes negligible when all labels are used, and it carries substantial latency and storage cost. Future work should stabilize the raw-scale features, add noise-aware training, evaluate independent multi-site data, incorporate random-kernel features, and compress the model for edge deployment.

Ethical statement and conflict of interest

This study uses anonymized device-level charging power sequences and operational status labels. It contains no human participants, human biological material, personally identifying information, or animal experiments; institutional ethics approval was therefore not applicable. The authors declare no conflict of interest.

Funding
This research was funded by the Scientific Research Fund of Hangzhou Dianzi University Information Engineering School, grant number KYP0223008.

Author contributions
Zhuotao Qian: methodology, software, formal analysis, investigation, data curation, visualization, and writing–original draft. Chenghua Feng: conceptualization, validation, resources, supervision, and writing–review and editing. Both authors approved the submitted manuscript.

Data availability
The charging records are not publicly available because they contain proprietary operational data supplied by an industrial platform. Access requests may be directed to the corresponding author and require approval from the data provider.

References

[1] Zhang J, Wang Y, Jiang B et al 2023 Realistic fault detection of Li-ion battery via dynamical deep learning *Nat. Commun.* **14** 5940

[2] Park S J, Kim W J, Kang B S et al 2022 Development of a fault-diagnosis system through the power conversion module of an electric vehicle fast charger *Energies* **15** 5056

[3] Duan Y, Shu S, Zhao Y et al 2024 Machine learning-based spatiotemporal fusion method for non-intrusive charging pile fault identification *Front. Electron.* **5** 1490939

[4] Chen W 2021 Research on time-series data classification methods based on deep learning Master's thesis (Wuhan: Huazhong University of Science and Technology) (in Chinese)

[5] Xu J, Hu W, Jin D et al 2026 Review of time-series data anomaly detection methods *Comput. Sci. Explor.* **20** 1517–1544 (in Chinese)

[6] Breiman L 2001 Random forests *Mach. Learn.* **45** 5–32

[7] Friedman J H 2001 Greedy function approximation: a gradient boosting machine *Ann. Stat.* **29** 1189–1232

[8] Cortes C and Vapnik V 1995 Support-vector networks *Mach. Learn.* **20** 273–297

[9] Bai S, Kolter J Z and Koltun V 2018 An empirical evaluation of generic convolutional and recurrent networks for sequence modeling arXiv:1803.01271

[10] Liu S, Sun J, Qin Y et al 2026 Time-series classification based on a multiscale Transformer and morphological convolution *Syst. Eng. Electron.* 1–14 (in Chinese)

[11] Yang Y, Hu K, Huang Y et al 2026 Classification of VLF/LF lightning electric-field time series using an improved Transformer *High Volt. Eng.* 1–16 (in Chinese)

[12] Jiang T 2026 Research on multivariate time-series forecasting and classification based on an improved Transformer Master's thesis (Huainan: Anhui University of Science and Technology) (in Chinese)

[13] Dempster A, Petitjean F and Webb G I 2020 ROCKET: exceptionally fast and accurate time series classification using random convolutional kernels *Data Min. Knowl. Discov.* **34** 1454–1495

[14] Dempster A, Schmidt D F and Webb G I 2021 MiniRocket: a very fast (almost) deterministic transform for time series classification *Proc. 27th ACM SIGKDD Conf. on Knowledge Discovery and Data Mining* pp 248–257

[15] Tan C W, Dempster A, Bergmeir C et al 2022 MultiRocket: multiple pooling operators and transformations for fast and effective time series classification arXiv:2102.00457

[16] Wang S and Yang Z 2026 Transfer randomized neural networks and their learning method for time-series classification *Sci. Technol. Eng.* **26** 10050–10061 (in Chinese)

[17] Yue Z, Wang Y, Duan J et al 2022 TS2Vec: towards universal representation of time series *Proc. AAAI Conf. Artif. Intell.* **36** 8980–8987

[18] Goswami M, Szafer K, Choudhry A et al 2024 MOMENT: a family of open time-series foundation models *Proc. 41st Int. Conf. on Machine Learning* **235** 16115–16152

[19] Das A, Kong W, Sen R et al 2024 A decoder-only foundation model for time-series forecasting *Proc. 41st Int. Conf. on Machine Learning* pp 10148–10167

[20] Ansari A F, Stella L, Turkmen C et al 2024 Chronos: learning the language of time series arXiv:2403.07815

[21] Shao H, Wang J, Zhang J et al 2026 Large-model-driven general fault diagnosis of rotating machinery in multiple scenarios *J. Mech. Eng.* 1–11 (in Chinese)

[22] Zhao X, Li Z, Wang Z et al 2026 Deep slope-displacement prediction based on Time-LLM *Hunan Commun. Sci. Technol.* **52** 29–34 (in Chinese)

[23] Yang J 2026 Time-series forecasting through optimized input representations for large models Master's thesis (Chongqing: Chongqing University of Technology) (in Chinese)

[24] Nie Y, Nguyen N H, Sinthong P et al 2023 A time series is worth 64 words: long-term forecasting with Transformers *Proc. 11th Int. Conf. on Learning Representations*

[25] Kim T, Kim J, Tae Y et al 2022 Reversible instance normalization for accurate time-series forecasting against distribution shift *Proc. Int. Conf. on Learning Representations*

[26] Saito T and Rehmsmeier M 2015 The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets *PLoS One* **10** e0118432

[27] Guo C, Pleiss G, Sun Y et al 2017 On calibration of modern neural networks *Proc. 34th Int. Conf. on Machine Learning* **70** 1321–1330